

Chapter 1 checkpoint!

Hello and welcome to your first checkpoint! Here come five questions, one about each of the learning targets from Chapter 1. This is your scorecard:

Learning target:	DF1	DF2	DFa	DFb	AD2
Your confidence level before starting (0-5):					
Your confidence level after the quiz (0-5):					
The mark you earned on this attempt:	Success! Try again!	Success! Try again!	Success! Try again!	Success! Try again!	Success! Try again!

Before anything else, please do the following:

- Rank your confidence from 0-5 on each of the learning targets. 5 means “I could teach a whole class about this;” 0 means “I am genuinely not sure I have heard these words before.”
- Write your name on this page and on each of the other pages of the quiz.

Then do the quiz! Some reminders:

- Open notes, closed computer.
- If you need more room to write, use the back of the same learning target page, or ask me for some scratch paper.
- Read the questions carefully and make sure you’re answering each part.
- Show all your work and explain all your thinking!

When you are done:

- Rank your confidence from 0-5 on each of the learning targets. 5 means “I absolutely nailed that question for sure;” 0 means “oof, I definitely didn’t get that one.”
- Make double sure your name is on every page, including any scratch paper.
- Hand in your work, separated by learning target.

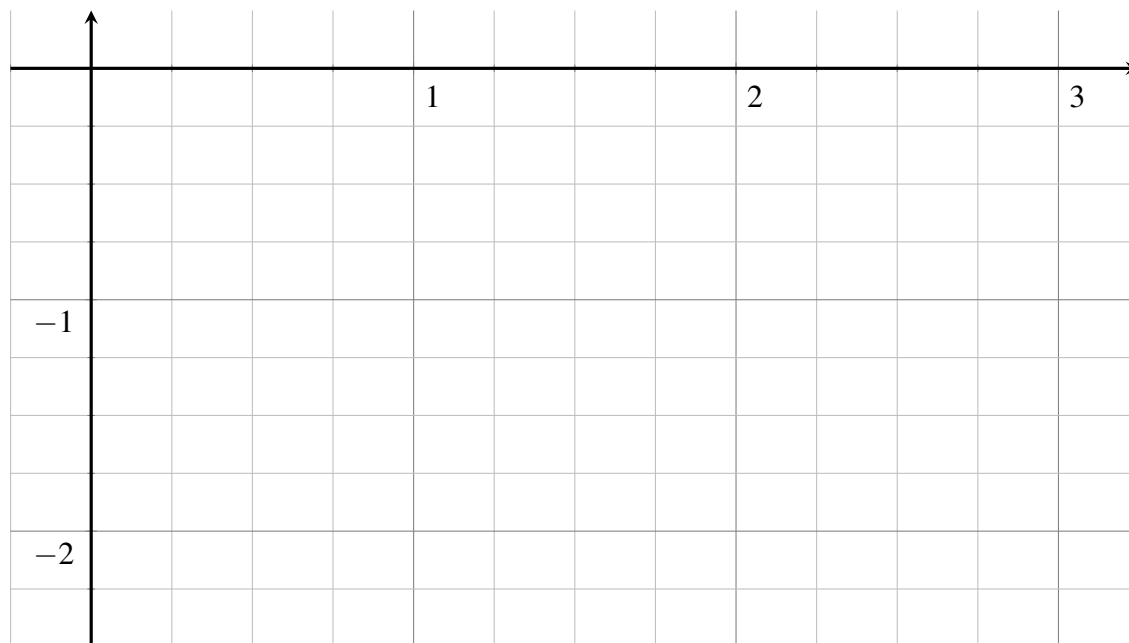
Have fun and do your best! I believe in u ♥

Learning target DF1, version 3/30

Suppose that you know the following values of some function $g(x)$:

x	1.7	2	2.3
$g(x)$	-2.2	-2	-1.5

1. Plot these points and sketch a plausible graph of $g(x)$.



2. On your graph above, draw a plausible tangent line to the graph of $g(x)$ at $x = 2$.
3. Use a central difference to estimate $g'(2)$. Draw the corresponding secant line on your graph above.
4. Compute another, different estimate of $g'(2)$. Draw the corresponding secant line on your graph.
5. Which one of your approximations is the best? How do you know?

Learning target DF2, version 3/30

Suppose that $p(x) = 4x^2 - 2x + 7$. Use the limit definition of the derivative to find $p'(x)$.

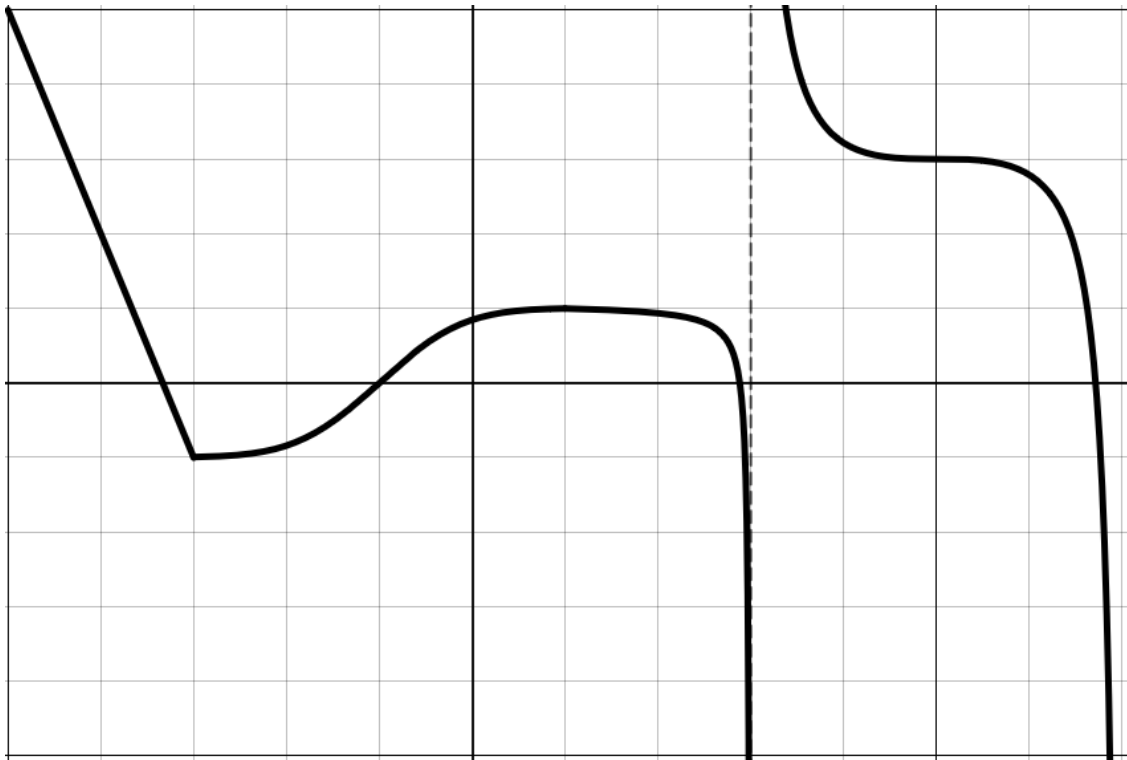
Learning target DFa, version 3/30

A company manufactures rope, and the total cost of producing r feet of rope is $C(r)$ dollars.

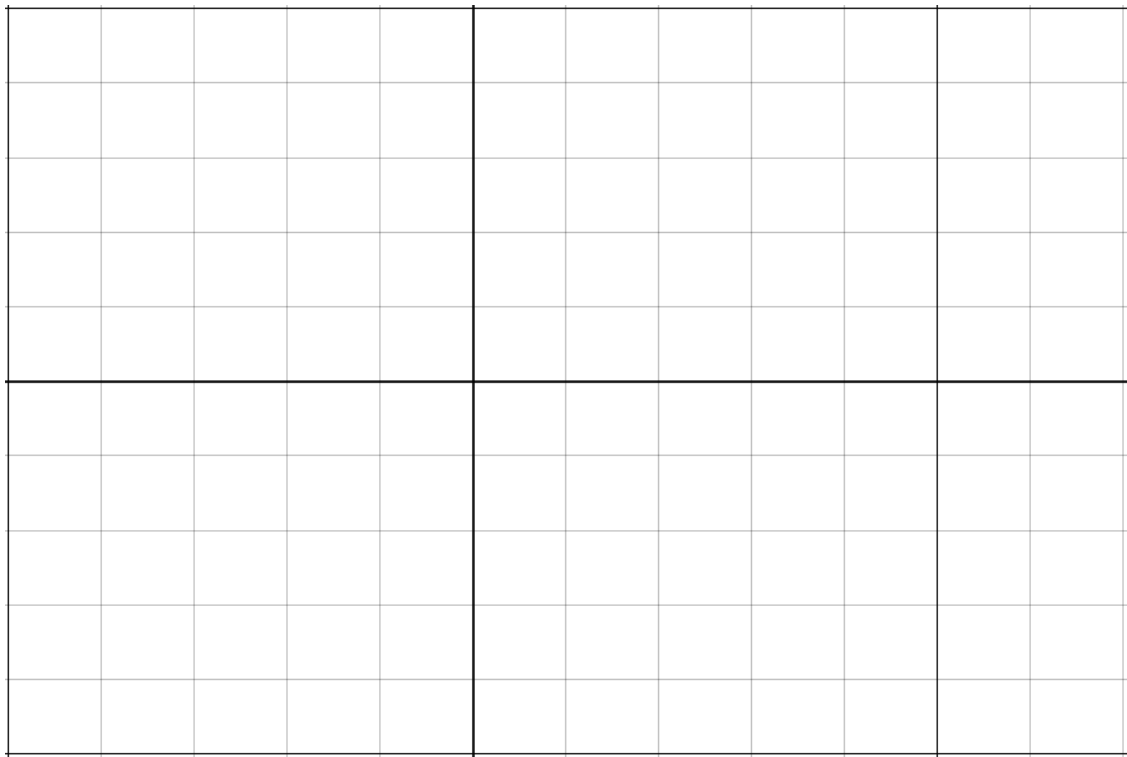
1. Suppose $C(2000) = 800$. Write a sentence explaining what this means, including units.
2. What are the units of $C'(r)$?
3. Suppose $C'(2000) = 0.35$. Write a sentence explaining what this means, including units.
4. Do you think $C'(3000)$ is greater than, equal to, or less than $C'(2000)$? Explain why.

Learning target DFb, version 3/30

Here is the graph of some wacky function $q(x)$:



Sketch the graph of $q'(x)$ on the blank axes below.



Learning target AD2, version 3/30

Suppose that for some function $p(x)$, you know the following information:

- $p(-2) = 5$,
- $p'(-2) = 1.5$,
- $p''(x) < 0$ for x -values close to -2 .

1. Explain and demonstrate how to find the linearization $L(x)$ of $p(x)$ at $x = -2$.
2. Explain and demonstrate how to estimate the value of $p(-2.03)$ using this linearization.
3. Is your estimate of $p(-2.03)$ greater than or less than the actual value? How do you know?
4. Sketch a possible graph of $p(x)$ and its linearization $L(x)$ nearby $x = -2$ to illustrate your findings. Label important points in your sketch with their coordinates.